

Mapping Best Empirical Models

A Meta-Analysis of Winning Solutions in Kaggle Tabular Competitions

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June 2026

CS495 Capstone in Data Science

Bellevue College

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Motivation

Winning a tabular Kaggle competition is widely taken to be about **elaborate feature engineering**.

Clever, hand-crafted features that set the winners apart.

Yet winners do modest feature engineering

Among the winners:

4

median own features
(best-documented winners)

6

winners engineered
no features

62%

simply ensembled
existing models

Feature-engineering volume looks less decisive than the conventional view suggests.

What we “know” is anecdotal

- Folk wisdom lives in blog posts, forum threads, and winner interviews.
- Academic benchmarks compare **algorithms** on fixed datasets, rarely **winners** against strong non-winners.
- Nobody has systematically asked what *distinguishes* a winning solution from a near-winning one.

This study codes 45 winning solutions + 11 near-winning controls.

Research questions

- RQ1** What strategy **paradigms** define winning? Do folklore “constraint → strategy” rules hold?
- RQ2** Which feature-engineering techniques do winners use, and how much?
- RQ3** Does feature engineering **distinguish** winners from strong non-winners?
- RQ4** How is the winning **population** structured?

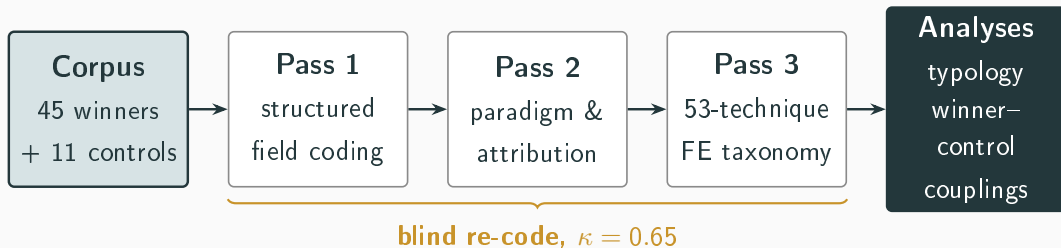
Data & Method

The corpus

- 45 winning solutions (top-three finishers)
- 11 near-winning controls (ranks 4–25, same competitions)
- February 2022 – April 2026
- Mostly Kaggle Playground Series (synthetic data); one real-data Featured competition
- Coded on a 40-column schema plus a 53-technique feature-engineering taxonomy
- Caveat: only solutions with public write-ups can be coded (documentation self-selection)

A content-analysis meta-study

No model is trained: the unit of analysis is **one completed solution**, coded in three passes.



Each winner is paired against a near-winning control from the *same* competition, isolating what distinguishes winning from merely strong play.

Checking our own coding

- Primary coding was *AI-assisted, researcher-directed*.
- A *blind re-code* of a stratified 12-entry subsample gives $\kappa = 0.65$: substantial agreement at the technique-family level.
- So FE results are reported at the *family* level, where they are reliable.

Limitations: small n (11 paired competitions); synthetic-data corpus; intra-rater (not inter-rater) reliability; documentation self-selection.

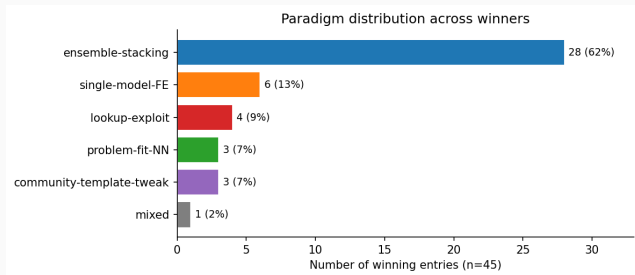
Findings

What winners actually use



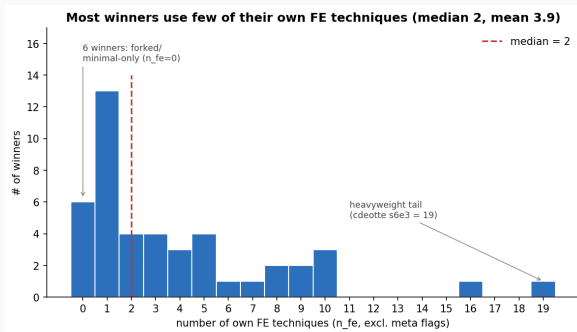
A few models (GBMs) and ensemble stacking dominate; FE is a long, varied tail.

Winning reduces to a few paradigms



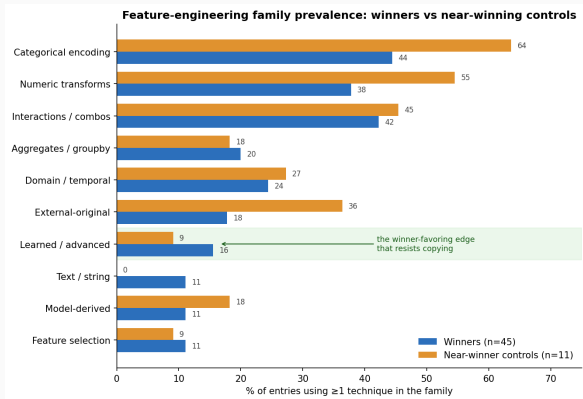
- **Ensemble stacking** = 62% of wins (28/45)
- Gradient-boosted trees are the dominant model in **35 of 45**
- A short tail: single-model-FE, lookup-exploit, problem-fit-NN

Winning FE is a small shared core



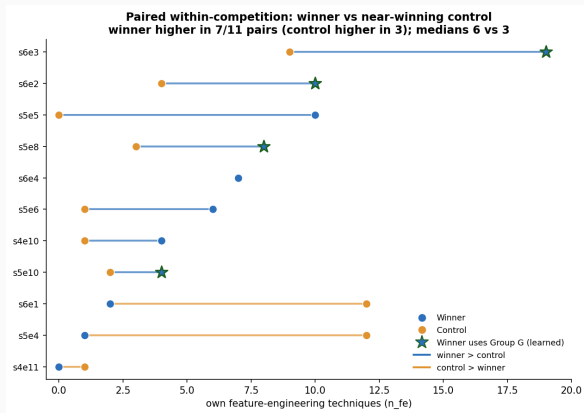
Even the best-documented winners engineer a median of 4 of their own features (a corpus-wide floor of 2), and 6 used none. Target encoding is the one near-universal technique; the rest is a long tail.

Aggregate FE: controls look like they do *more*



- Pooled across all entries, controls match or beat winners on nearly every FE family.
- But this is a **confound**: the 11 competitions that have codeable controls are FE-heavy (winner median $n_{fe} = 6$ vs. corpus-wide 2).
- So the aggregate measures a **competition-mix** difference — not a winner-vs.-control one.

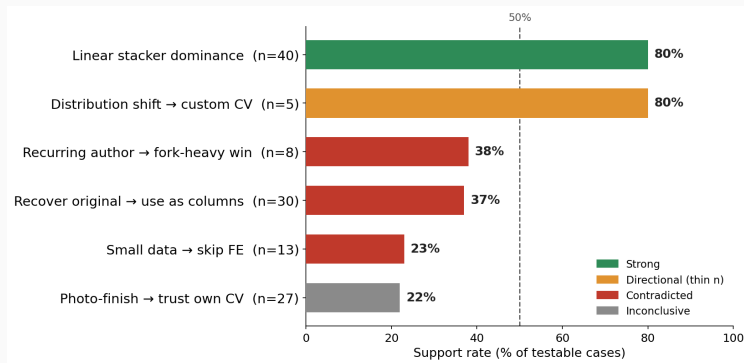
Paired within-competition: the gap reverses (still n.s.)



- Each control vs. the winner of its *own* competition removes the mix confound.
- Winners do more in 7 of 11 pairs; medians 6 vs. 3.
- Modest, **not significant** ($n = 11$, $p \approx 0.34$) — a directional correction.
- **Learned features** (Group G) lean winner, *never* controls — the edge that resists copying.

Either framing: FE volume does not **cleanly separate** winners from near-winners.

Folklore “rules” mostly don’t hold

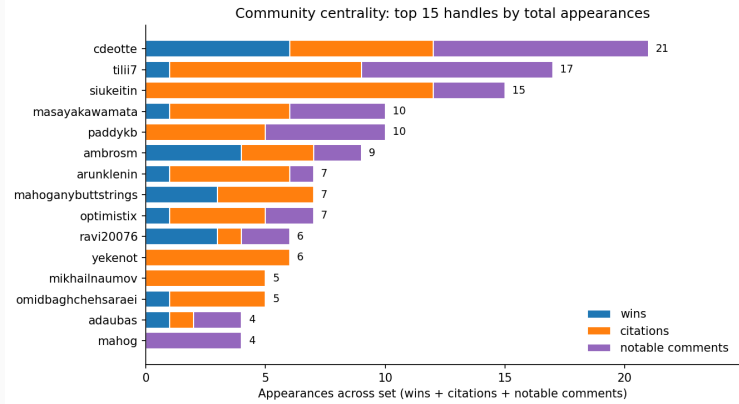


The lone green survivor, **linear-stacker dominance**, just restates standard ensembling.
The 80% “custom CV” rule rests on only **5 cases**.

The documented top is a recurring guild

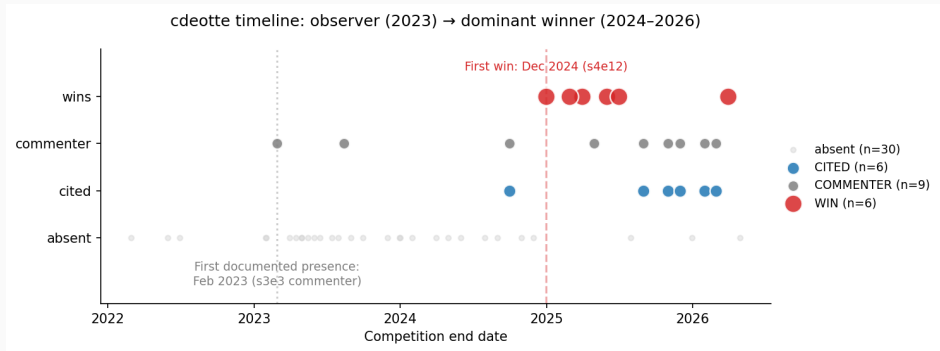
45%

author overlap between
winners and near-winners



A few central figures supply widely-used techniques without always winning (originators vs. canonizers); recognition accrues to the already-prominent.

How membership forms



- The most central figure: **observer in Feb 2023** → **six wins** by Mar 2026.
- Nearly **two years** of comments, citations, and non-winning entries *before* the first win.
- Recognition follows **participation**, not just skill — the *order* of events is the point. 15/18

Conclusions

Three takeaways

1. Feature engineering is a **small shared core**, not the elaborate craft commonly assumed.
2. **Learned features** are the one feature-level edge that resists copying (directional, on small counts).
3. Winning is, in part, a **community phenomenon**.

What this does, and does not, establish

- Claims are scoped to the **documented competitive elite**: not typical competitors, not real-world tabular ML.
- The corpus is mostly **synthetic** Playground data; some strategies are generator artifacts.
- The winner-control test is **underpowered** ($n = 11$): the result is “no clean separation,” not “no difference.”

Contributions & future work

Contributions

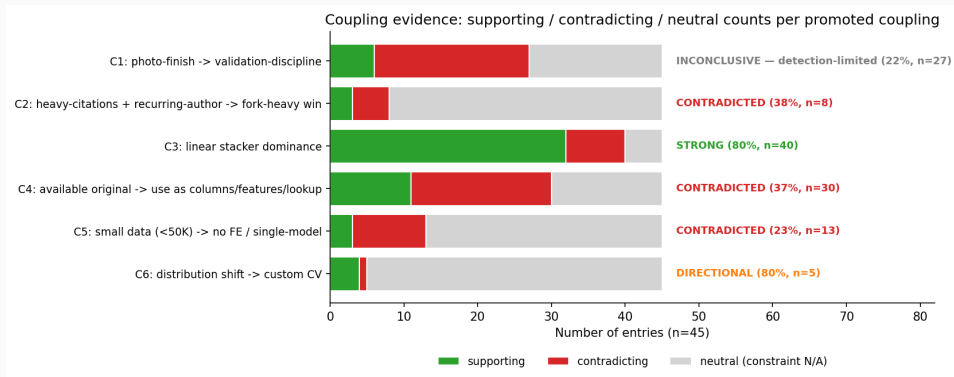
- The first structured, reliability-checked characterization of winners *versus near-winners*.
- A reusable 53-technique FE taxonomy and multi-pass coding workflow.
- Replaces folklore with documented frequencies.

Future work

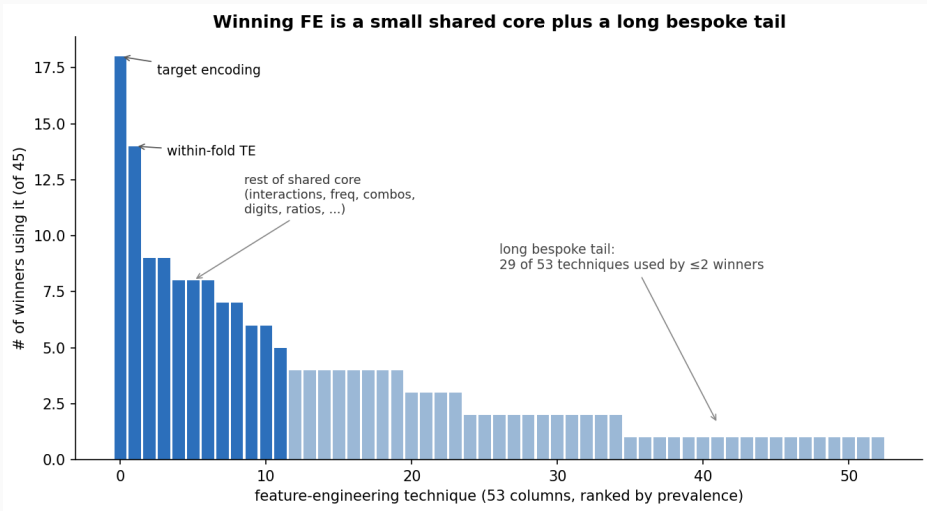
- Track the emerging **Season-6 shift**: very large (100+ model) ensembles and LLM-assisted workflows the four-paradigm typology will need to absorb.
- Real-data Featured competitions; other platforms and modalities.

Winning turns less on the *volume* of feature engineering
than on **paradigm fit** and **who competes**.

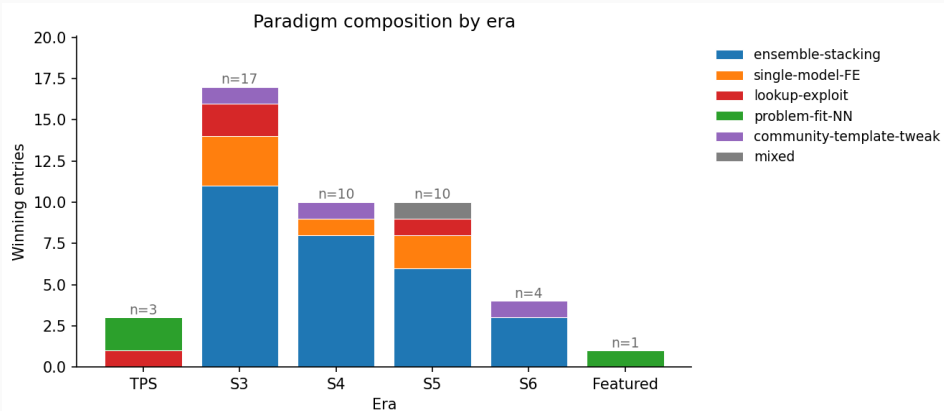
Appendix: coupling evidence (detailed)



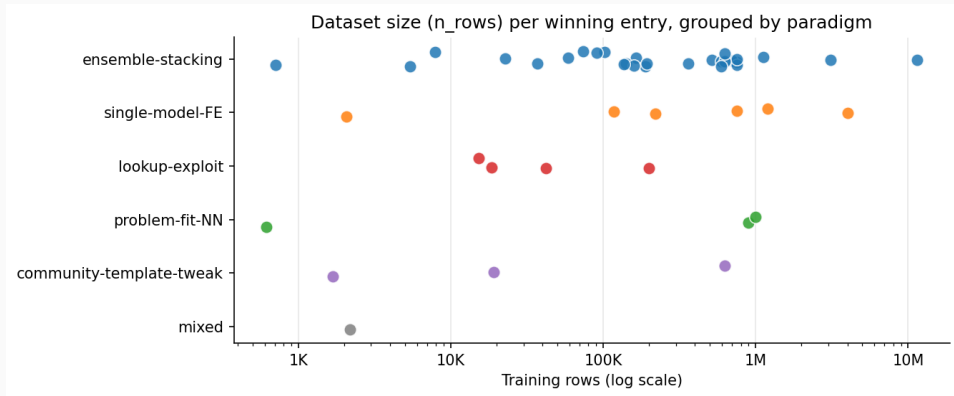
Appendix: feature-engineering technique long tail



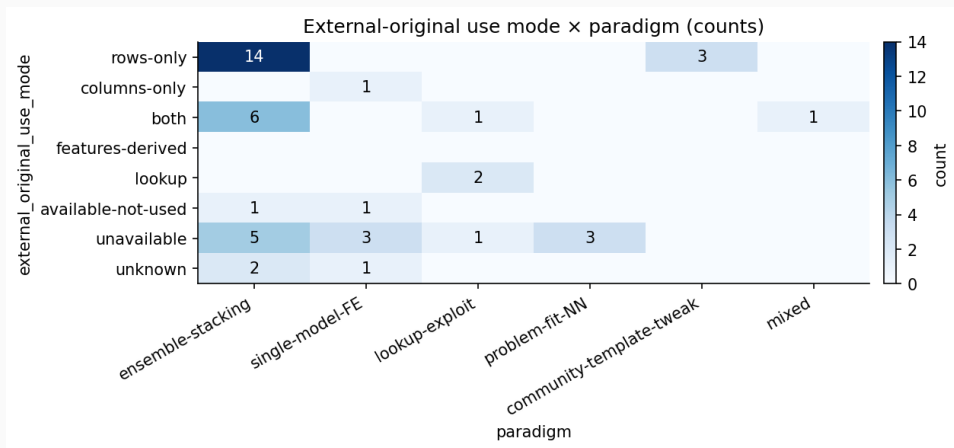
Appendix: paradigm composition by era



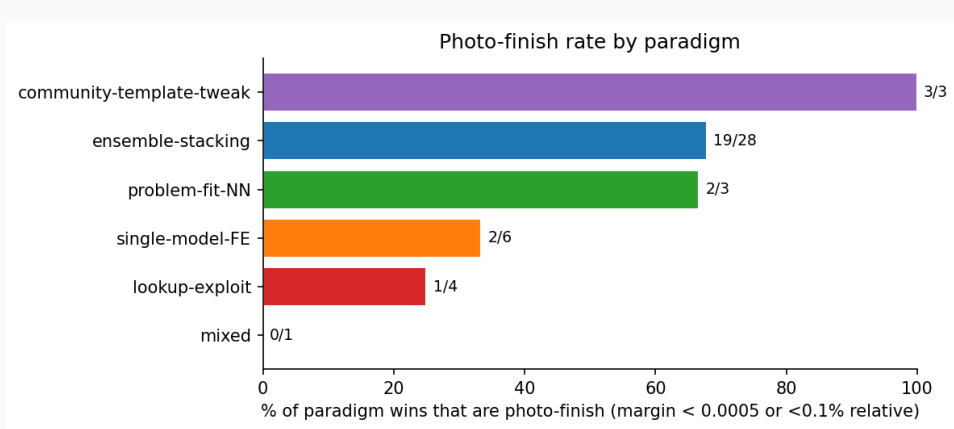
Appendix: dataset size by paradigm



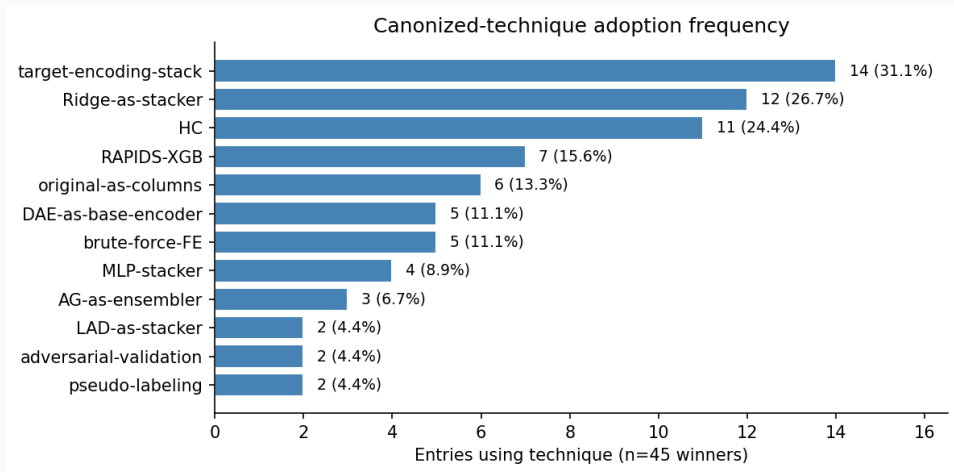
Appendix: external-original use by paradigm



Appendix: photo-finish rate by paradigm



Appendix: most-canonized techniques



Appendix: citations by origination score

